PoMaS 譁堺ｽ懊・繝九Η繧｢繝ｫ v0.92

**M5Stack Cardputer-Adv 逕ｨ 螟壽ｩ溯・謨ｰ蟄ｦ繧ｱ繝・・繧ｷ繝ｧ繝ｳ 蝗樊・迚・*

逶ｮ谺｡

1. 縺ｯ縺倥ａ縺ｫ
2. 蛛穂ｽ懆腸蠅・・繧､繝ｳ繧ｹ繝医・繝ｫ
3. 蟇ｾ諠ｬ譁堺ｽ・
4. Calculator 繝｢繝ｼ繝・
5. CAS 繝｢繝ｼ繝・
6. Statistics 繝｢繝ｼ繝・
7. Scientific Reference 繝｢繝ｼ繝・
8. Graph 繝｢繝ｼ繝・
9. Tools 繝｢繝ｼ繝・
10. 險ｭ螳夲ｼ・VS 豌ｸ邯壼喧・・
11. 繝医Λ繝悶Ν繧ｷ繝･繝ｼ繝・ぅ繝ｳ繧ｰ
12. 莉倬鍬・磯未謨ｰ荳☒隕ｧ / 繧ｭ繝ｼ謫ｰ隕玖｡ｨ・・

1. 縺ｯ縺倥ａ縺ｫ

PoMaS・・ocket Math Station・峨・縲｀ 5Stack Cardputer-Adv 荳翫〒蜍穂ｽ懊☆繧句､壽ｩ溯・髢｢謨ｰ髮ｻ蜊薙〒縺吶☒・

謨ｰ蠑剰ｩｕ萓｡蜉・CAS繝ｻ邨ｱ險郁ｧ｣譫舌・繧ｰ繝ｩ繝墓緒逕ｻ繝ｻ邁｡譏薙Ξ繧ｹ繝・ｰ譛ｭ蜷医∴繝ｻ繧ｻ繝ｳLisp 繧､繝ｳ繧ｿ繝励Μ繧ｿ・1 蜿ｰ縺ｫ邨ｱ蜷医＠縺ｦ縺・∪縺吶☒・

險ｭ險域☒晄Φ・・

- **1 逕ｻ髱｢ 1 讖滓・** 蠖・諠・☒ｱ驕主、壹ｒ驕ｿ縺代☒∬ｿｷ繧上↑縺・UI
- **繧ｭ繝ｼ繝懊・繝牙ｮ檎ｵ・* 蠖・繧ｿ繝・メ荳崎ｦ√☒∝・縺ｦ縺ｮ譁堺ｽ懊′ 迚ｩ逅・く繝ｼ縺ｧ螳檎ｵ・
- **Paper UI v2** 蠖・豺｡繝溘Φ繝育ｴ呵レ譎ｯ ＋ 繧ｰ繝ｩ繝輔ぅ繧､繝域枚蟄励〒髫ｷ譁る侯菴懈･ｭ縺ｫ螟ｩｪ縺励＞
  ＞

2. 蜍穂ｽ懃腸蠅・・繧､繝ｳ繧ｹ繝医・繝ｫ

2.1 繝上・繝峨え繧ｧ繧｢隕ａｻｶ

鬆・峅	隕ａｻｶ
繝・ヰ繧､繧ｹ	M5Stack Cardputer-Adv・・SP32-S3FN8・・
Flash	8 MB
PSRAM	髢楚政霈会ｼ・00 隕∫ｴ☒縺ｾ縺ｧ縺ｮ邨ｱ険医Ｍ繧ｹ繝茨ｼ・
逕ｻ髱｢	ST7789V2 240ﾃ・35 px
SD 繧ｫ繝ｼ繝・	莉ｻ譁擾ｼ・AT32, 32GB 莉･荳区耳螂ｨ・・

2.2 繝輔ぃ繝ｼ繝☒繧ｦ繧ｧ繧｢譖ｸ縺崎ｾｼ縺ｿ

1. esptool 繧偵う繝ｳ繧ｹ繝医・繝ｫ
2. Releases 縺九ｉ `PoMaS_v0.92_Cardputer-Adv.bin` 繧偵ム繧ｦ繝ｳ繝ｭ繝ｼ繝・
3. Cardputer-Adv 繧・USB-C 縺ｧ謗･邯・
4. 譖ｸ縺崎ｾｼ縺ｿ・・

```

esptool.py --chip esp32s3 --port COM3 --baud 921600 \
  write_flash --flash_mode dio --flash_freq 80m --flash_size 8MB \
  0x0 PoMaS_v0.92_Cardputer-Adv.bin
```

`COM3` 縺ｯ迺ｰ蠅・↓蜷医ｏ縺帙※螟画峩縺励※縺上□縺輔＞縲ゅ↑縺後ｂ縺ｨ縺ｯ繧ｨ繝ｼ繝ｩ繝ｼ縲ヨ縺ｧ繝☒繧ｦ繝ｳ繝ｭ繝ｼ繝峨Δ繝ｼ繝峨↓蜈･繧後ｌ縺吶☒・

2.3 襍ｷ蜍・

繝ｪ繧ｻ繝・ヨ 竊・繧ｹ繝励Λ繝・す繝・・育ｴ・2.5 遘抵ｼ俄・ 繝｡繧､繝ｳ繝｡繝九Η繝ｼ

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3. 蝓ｺ譛ｬ謫堺ｽ・

3.1 蜈ｱ騾壹く繝ｼ

繧ｭ繝ｼ	**蜍穂ｽ・**
`Fn + Esc`・医ヰ繝・け繧ｯ繧ｩ繝ｼ繝茨ｼ・	謌ｻ繧・/ 繧ｭ繝｣繝ｳ繧ｻ繝ｫ
`Enter`	豎ｺ螳・/ 螳溯｡・
`Del`	繝舌ャ繧ｯ繧ｹ繝壹・繧ｹ
`Fn + 謨ｰ蟄輿	蜷・Δ繝ｼ繝峨・繧ｷ繝ｧ繝ｼ繝医き繝・ヨ
`;`	荳・
`.`	荳・
`,`	蟾ｦ
`/`	蜿ｳ

3.2 繝｡繧､繝ｳ繝｡繝九Η繝ｼ

1. **Calculator** 窶・謨ｰ蠑城崕蛹・
2. **CAS** 窶・謨ｰ蠑丞・逅・
3. **Statistics** 窶・邨ｱ險郁ｧ｣譫・
4. **Scientific Reference** 窶・遘大ｭｦ雉・侭
5. **Graph** 窶・繧ｰ繝ｩ繝墓緒逕ｻ
6. **Tools** 窶・繧ｹ繝励Ξ繝・ラ繧ｷ繝ｼ繝・/ ﾎｼLisp / Wi-Fi
7. **About** 窶・繝舌・繧ｸ繝ｧ繝ｳ諠・☒ｱ

4. Calculator 繝｢繝ｼ繝・

4.1 蝓ｺ譛ｬ險育ｮ・

蠑上ｒ蜈･蜉帙＠縺ｦ `Enter` 縲ょｯｾ蠢懈ｧ区枚・・

貍皮ｮ怜ｭ・/ 髢｢謨ｰ	**萓・**
蝗帛援貍皮ｮ・	`2+3*4` 竊・14
縺ｹ縺堺ｹ・	`2^10` 竊・1024
髫惹ｹ・	`5!` 竊・120
蜑ｰ菴・	`10 mod 3` 竊・1
蟷ｳ譁ｹ譬ｹ	`sqrt(2)`
荳芽ｧ帝未謨ｰ	`sin(pi/2)`, `cos(60)`, `tan(45)`
蟇ｾ謨ｰ	`ln(e)`, `log(100)`, `log2(8)`
邨・粋縺・	`nCr(5,2)`, `nPr(5,2)`
邏☒謨ｰ蛻､螳・	`isPrime(97)`
譁・ｭ怜ｮ壽焚	`pi`, `e`, `euler`
螟画焚	`A`縲象J`, `E` 縺ｯ螟画焚縺ｨ縺励※菴ｿ逕ｨ蜿ｯ
逶ｴ蜑阪・邨先棡	`Ans`

4.2 繧ｷ繝ｧ繝ｼ繝医き繝・ヨ

繧ｭ繝ｼ	**讖溯・**
`Fn+1`	險ｭ螳夲ｼ郁ｧ貞ｺｦ繝ｻ陦ｨ遉ｺ蠖｢蠑上・譯∵焚・・
`Fn+2`	螻･豁ｴ蜻ｼ縺ｳ蜃ｺ縺暦ｼ域怙螟ｧ 100 莉ｶ・・
`Fn+3`	繝｡繝｢繝ｪ M・・+ / M- / MR / MC / Set・・
`Fn+4`	螟画焚 A縲廱
`Fn+5`	蝓ｺ謨ｰ螟画鋤・・EC / HEX / OCT / BIN・・
`Fn+6`	繝薙ャ繝域ｼ皮ｮ暦ｼ・ND / OR / XOR / NOT / SHL / SHR・・
`Fn+7`	隍・ｴ☒謨ｰ貍皮ｮ暦ｼ亥ｮ溯ｻｸ / 讌ｵ蠖｢蠑・/ 髢｢謨ｰ・・
`Fn+8`	謚｡蝶ｵ繝・・繝ｫ・井ｸ玖ｨ假ｼ・
`Fn+9`	RPN 繝｢繝ｼ繝牙・譖ｿ
`Fn+0`	繝倥Ν繝・
`Fn+=`	蛻・焚 卄・蟆乗焚陦ｨ遉ｺ
`Fn+a`	Ans 繧貞ｼ上↓謖ｿ蜈･
`Fn+b`	Ans 繝舌Φ繧ｯ・域怙螟ｧ 20 莉ｶ・・

4.3 謚｡蝶ｵ繝・・繝ｫ・・n+8・・

識溯・	蜀・ｮｹ
譁ｹ遖句ｼ上た繝ｫ繝舌・	Newton / 莠悟・豕・/ 2縲・ 谺｡譁ｹ遖句ｼ・/ 驟｣遶・1 谺｡譁ｹ遖句ｼ・/ 荳☒閭ｬ螟夐☒・ｼ擾ｼ・eg 竕、8・・
謨ｰ蛟､隗｣譫・	蠕ｮ蛻・/ 遨榊・ / 邱丞柱 / 邱丈ｹ・
繝吶け繝医Ν繝ｻ陦悟・	繝峨ャ繝育ｩ・/ 繧ｯ繝ｭ繧ｹ遨・/ 繝弱Ν繝☒ / 蜉☒貂帷ｮ・/ 陦悟・蟄礼ｮ・/ 陦悟・蠑・/ 驟・｡悟・ / 霆｢鄂ｮ / 髻取焚 / 蝗ｺ譛牙☒、
蠏ｩ蓮包ｼ・7 蝗ｳ蠖｢・・	荳芽ｧ貞ｽ｢ 3 遞ｮ / 蝗幄ｧ貞ｽ｢ 5 遞ｮ / 蜀・/ 謇・ｽ｢ / 譎募・ / 迺− / 豁｣螟夊ｧ貞ｽ｢ / 遶倶ｽ・8 遞ｮ / 豁｣螟夐擇菴・5 遞ｮ / 繝医・繝ｩ繧ｹ
遖ｺ邇・す繝溘Η繝ｬ繝ｼ繧ｷ繝ｧ繝ｳ	繧ｳ繧､繝ｳ / 繧ｵ繧､繧ｳ繝ｭ / 繧ｬ繝√Ε / 蒸ｱ譁− / 豁｣隕丈ｹｱ譁− / 謖・焚蒸ｱ譁− / 隱ｬ 函謁･繝代Λ繝峨ャ繧ｯ繧ｹ
邏☒蝗☒謨−蛻・ｧ｣	64 繝薙ャ繝医∪縺ｧ縺ｮ謨ｴ謨−

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5. CAS 繝｢繝ｼ繝・

螟夐☒・ｼ上・險倅捷蠑ｧ逅・☒A x^3 - 2x + 1` 縺ｮ繧医≧縺ｪ蝶上ｒ 蜈･蜉☒・

識溯・	隱ｬ譏・
Eval at x	螟夐☒・ｼ上・ x 謖・ｮ夂ｩ穂ｾ｡
Differentiate	1 髫主ｾｮ蛻−
Integrate	荳榊ｮ夂ｩ榊・・・ C・・
Add	2 螟夐☒・ｼ上・蜥・
Multiply	2 螟夐☒・ｼ上・遨・
Roots	譬ｹ・郁､・ｴ☒謨−蟇ｾ蠢懶ｼ・
Taylor	莉ｻ諢城未謨−縺ｮ謨−蛟､繝・う繝ｩ繝ｼ螻暮幕・域ｬ｡謨− 1縲・・・
Limit	謨−蛟､譛ｫ髯撰ｼ井ｸ｡蛛ｴ譛ｫ髯舌・荳☒闘ｴ蛻､螳夲ｼ・
Substitute	險倅捷鄂ｮ謠幢ｼ井ｾ・ `x` 竊・`y+1`・・

螟夐☒・ｼ上・譖ｸ蝶丈ｾ具ｼ・

- `1+2x-x^2`
- `3x^2-2x+1`
- `x^3-x`

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6. Statistics 繝｢繝ｼ繝・

6.1 Lists & Descriptive

- L1縲廰8 縺ｮ 8 繝ｪ繧ｹ繝医⊂∝推 200 隕∫ｴ⊂・・SRAM 謳ｭ霈画凾 999・・
- 譎塀ｽ懶ｼ啻Enter`・育ｷｨ髪・ｼ・`Del`・亥炎髯､・・`S`・医た繝ｼ繝茨ｼ・`A`・郁ｿｽ蜉⊂・・
- 霑ｽ蜉⊂譎ゅ・繧ｫ繝ｳ繝槫玄蛻・ｊ縺ｧ隍・焚蛟､蜈･蜉帛庄・啻1, 2, 3, 4`
- 險倩ｿｰ邨ｱ險医・繝ｼ繧ｸ 1・嗜, sum, sum2, mean, sd(samp), sd(pop), var(samp), SE, CV, min, max
- 險倩ｿｰ邨ｱ險医・繝ｼ繧ｸ 2・啣1, med, Q3, IQR, skew, kurt, mode

6.2 Variable Scale

蜷・M繧ｹ繝医↓蝪ｺ蟒ｺ繧定ｨｭ螳夲ｼ壼ｯｫ比ｾ・/ 髢馴囈 / 鬆・ｺ・/ 蜷咲ｾｩ

6.3 迚ｹ谿雁ｹｳ蝮・

- 驥阪∩莉倥″ 蟷ｳ蝮・ｼ・1=蛟､, L2=驥阪∩・・
- 蟶ｾ菴募ｹｳ蝮・
- 隱ｿ蜥悟ｹｳ蝮・
- 繝医M繝⊂蟷ｳ蝮・ｼ・lpha 0縲・.5・・

6.4 Frequency Distribution

繝薙Φ謨ｰ–譚・ｮ壹〒繝偵せ繝医げ繝ｩ繝⊂ + 鬆ｻ蠎ｦ陦ｨ繧定。ｨ遉ｺ縲・

6.5 Z-Scores

譚・ｮ壹M繧ｹ繝医・ Z-score 繧貞挨繝ｪ繧ｹ繝医↓菫晏ｭ倥⊂・

6.6 Regression・亥肓蠼ｰ・・

遮ｮ鬘・	**蝶・**
Linear	y = a + bx
Quadratic	y = a + bx + cxﾂｲ
Power	y = a ﾂｷ x^b
Exponential	y = a ﾂｷ e^(bx)
Logarithmic	y = a + bﾂｷln(x)
Logistic	logit(y) = a + bx
Med-Med	荳ｭ螟ｮ蛟､繝吶・繧ｹ縺ｮ邱壼ｽ｢蝗槫ｸｰ
Multiple	螟夐瑚邱壼ｽ｢蝗槫ｸｰ・・ 竕､5・・
Stepwise	螟画焚蠅怜刈豕輔↓繧医k 螟画焚驕ｸ諞・

蚤ｺ蛉幢ｼ啾, b, c, r, rﾂｲ, 莠域ｸｬ蛟､繧よｱ句ｷｮ縺ｮ菫晏ｭ倥 b 蜿ｯ閭ｽ繧・

6.7 Tests・井ｻｮ隱ｬ讀懷ｮ夲ｼ・

讀懷ｮ・	**蜀・ｮｹ**
z-test (1-sample)	ﾏ・譌｢遏･縺ｮ 1 讓呎悽讀懷ｮ・
t-test (1-sample)	1 讓呎悽 t 讀懷ｮ・
t-test (2-sample, eq.)	遲牙・謨｣ 2 讓呎悽 t 讀懷ｮ・
Welch's t-test	荳咲ｭ牙・謨｣ 2 讓呎悽 t 讀懷ｮ・
Paired t-test	蟇ｾ蠢懊≠繧・t 讀懷ｮ・
F-test	蛻・淵豈疲､懷ｮ・
One-way ANOVA	荳⊠蜈・・鄂ｮ蛻・淵蛻・梵
Two-way ANOVA	莠悟・驟咲ｽｮ蛻・淵蛻・梵
Chi-Square GOF	驕ｩ蜷亥ｺｦ讀懷ｮ・
Chi-Square Independence	辺ｬ遶区⊠ｧ讀懷ｮ夲ｼ・ﾃ幼・・
Fisher's exact (2ﾃ・)	繝輔ぅ繝・す繝｣繝ｼ豁｣遒ｺ讀懷ｮ・

蚤ｺ蛉幢ｼ壽､懷ｮ夂ｵｱ險磯㍼繝・・f繝｝　蛟､繝∵快譚丞愛螳夲ｼ夷ｱ=0.05・・

6.8 Non-parametric / Risk

識溯・	**蜀・ｮｹ**
Mann-Whitney U	繝弱Φ繝代Λ 2 郛､豈碑ｼ・
Wilcoxon signed rank	蟇ｾ蠢懊≠繧翫ｯ繝ｳ繝代Λ讀懷ｮ・
Kruskal-Wallis	繝弱Φ繝代Λ 3 郛､莉･荳・
Friedman	蜿榊ｾｩ貂ｬ螳壼ｯ繝ｳ繝代Λ
Sign test	隨ｦ蜿ｷ讀懷ｮ・
Spearman correlation	鬆・ｽ咲嶌髢｢
Kendall tau	鬆・ｽ咲嶌髢｢・医こ繝ｳ繝峨・繝ｫ・・
Odds Ratio + 95% CI	繧ｪ繝・ぜ豈・
Risk Ratio + 95% CI	繝ｪ繧ｹ繧ｯ豈・
Risk Difference + 95% CI	繝ｪ繧ｹ繧ｯ蟾ｮ
NNT	豐ｻ療ｮｿ・ｦ∵焚
CI (mean)	蟷ｳ蝮・・菫｡鬆ｼ蛹ｺ髢・
CI (proportion, Wilson)	豈皮紫縺ｮ菫｡鬆ｼ蛹ｺ髢・

6.9 Distributions・亥・螳・ｹ繝ｩ繧ｦ繧ｶ・・

- 豁｣隕・N(ﾎｼ, ﾏ・
- t 蛻・ｸ・ｼ・f・・
- ﾏ・ｲ 蛻・ｸ・ｼ・f・・
- F 蛻・ｸ・ｼ・1, d2・・
- 莠基☒・ｼ・, p・・
- 繝昴い繧ｽ繝ｳ・夷ｻ・・
- 蟷ｾ菴包ｼ・・・

蜷・・螳・〒・唔MF/PDF縲，DF縲・☒・未譲–縲￣DF 繝励Ο繝・ヨ

6.10 Multivariate・亥、壷、蛾蹣隗｣譫撰ｼ・

識湖・	蜀・ｮｹ
PCA	荳ｻ諤仙・蛻・梵・亥華譖牙☒､縺ｻ蟇・ｸ守紫縺ｻ雋☒闕ｷ驥擾ｼ・
Factor Analysis	蝗☒蟄仙・譫撰ｼ・arimax 蝗樊ｻ｢・・
k-means	k 蠑ｳ蝮・け縺ｩ繧ｹ繧ｿ繝ｪ繝ｳ繧ｰ
Hierarchical (Ward)	髫主ｱ､繧ｯ繝ｩ繧ｹ繧ｿ繝ｪ繝ｳ繧ｰ
Decision Tree	CART 蝗槫ｸｰ譛ｨ

6.11 Stratified・亥ｱ､蛻･隗｣譫撰ｼ・

- Mantel-Haenszel OR・・er-stratum + pooled + Forest plot・・
- Mantel-Haenszel RR
- Food Poisoning Trace・磯｣滉ｸｭ豈貞次蝗☒鬢溷刀迚ｹ螳夲ｼ・
- Cochran's Q・育焚雉ｪ諤ｧ讀懷ｮ夲ｼ・
- Breslow-Day 讀懷ｮ・

6.12 Survival・育函蟄伜・譫撰ｼ・

- Kaplan-Meier・育函蟄俶峇邱夲ｼ・
- Log-rank 讀懷ｮ・
- Cox 豈比ｾ九ワ繧ｶ繝ｼ繝会ｼ・ 蝨ｧ螟蛾蹣・・
- Median Survival・育函蟄倅ｸｭ螟ｮ蛟､・・

6.13 Meta-analysis・医Γ繧ｿ繧｢繝翫Μ繧ｷ繧ｹ・・

- Inverse variance・亥崋螳壼柑譫懈ｼ・
- DerSimonian-Laird・医Λ繝ｳ繝☒繝☒蜉ｹ譫懈ｼ・
- Forest plot
- Funnel plot
- Egger's test・亥・迚医ヰ繧､繧｢繧ｹ・・

6.14 Statistical Graphs・・7 遮ｮ・・

Histogram / Box-and-Whisker / Scatter / Q-Q / Bar / Pie / Line / Stem-and-leaf / Pareto / ECDF / KDE / Strip / Cumulative / MeanｱSD / Residuals / P-P / Multi-line

6.15 CSV Stream Stats

SD 繧ｵ繝ｼ繝峨・ CSV 繧定。悟腰菴阪〒繧ｹ繝医Μ繝ｼ繝溘Φ繧ｰ蛹ｦ逅・☒ょ・縺斐→縺ｮ邨ｱ險・+ 謖・ｮ・2 蛻励・蝗槫ｸｰ繧・

7. Scientific Reference 繝｢繝ｼ繝・

繧ｵ繝・ざ繝ｪ	蜀・ｹ
Constants・・86 莉ｶ・・	迚ｩ逅・/ 謨ｰ蟄ｦ / EE / 螟ｩ譁・/ 誼第弐 / 緋匆ヱ譏・/ 蝨ｰ逅・黄逅・/ 螟ｧ豌・/ 豬ｷ豢・/ 蜈・ｴ⊠ / 遥溽黄
Scientific Laws・・00 莉ｶ・・	迚ｩ逅・/ 蛹門ｭｦ / 謨ｰ蟄ｦ / 蝨ｰ逅・ｧ大ｭｦ / 螟ｩ譁・/ 遥溽黄 / EE
Unit Converter・・32 蜊倅ｽ・/ 20 繧ｵ繝・ざ繝ｪ・・	髟ｷ縺・/ 髱｢遨・/ 菴鍋ｩ・/ 雉ｪ驥・/ 驟溷ｺｦ / 蜉⊠驟溷ｺｦ / 蜉・/ 蝨ｧ蜉・/ 繧ｨ繝阪Ν繧ｮ繝ｼ / 莉穂ｺ狗紫 / 貂ｩ蠎ｦ / 隗貞ｺｦ / 譎る俣 / 諠・⊠ｱ驥・/ 蜻ｨ豕｢謨ｰ / 豬・蹴 / 蟇・ｺｦ / 邊倅ｺｦ / 譏ｾ蟆・ｷ・/ 縺昴・莉・
Periodic Table・・18 蜈・ｴ⊠・・	蜈・ｴ⊠險倅捷繝ｻ蝪溷ｭ宣匍繝ＡＹ` 縺ｧ Ans 縺ｫ霆｢驟√⊠～B` 縺ｧ Ans 繝舌Φ繧ｯ縺ｸ
Search by Symbol	險倅捷繝ｻ蜷榊燕縺ｧ蝗壽焚讀懃ｴ｢

8. Graph 繝｢繝ｼ繝・

8.1 繝｡繝九Η繝ｼ

鬆・岼	讖溯・
Plot current functions	迴ｾ蝨ｨ縺ｮ髢｢謨ｰ繧偵・繝ｭ繝・ヨ
Edit functions	諧⊠螟ｧ 4 諧ｨ縺ｮ髢｢謨ｰ邱ｨ髮・
Plot type	謠冗判繧ｿ繧､繝鈴∈謚・
Range / Window	遽・峇險ｭ螳・
Standard window	-10繝・0
Trig window	-2ﾏ⊠繝・ﾏ⊠
Square aspect (1:1)	逵溷・陦ｨ逶ｺ逕ｨ縺ｫ Y 遽・峇繧定・蜍戊ｪｿ謨ｴ
G-Solve	譬ｹ / 讌ｵ蛟､ / 莠､轤ｹ / 蝗ｺ夂榊・
Tangent / Normal	謗･邱壹・豕慕ｷ・
Save plot as BMP	BMP 菫晏ｭ・

8.2 Plot type

繧ｿ繧､繝・	蜈･蜉帑ｾ・
y = f(x)	`sin(x)`
Parametric	`cos(t);sin(t)` ・医そ繝溘さ繝ｭ繝ｳ蛹ｺ蛻・ j ・・
Polar	`1+cos(t)`
Implicit	`x^2+y^2-4`
Inequality	`x^2+y^2-1` ・・(x,y) > 0 鬆伜沺蝪励 j ・・
Sequence	`1/n` ・・ 繧剃ｽ迺ｨ・・
3D	`sin(x)*cos(y)`

8.3 繝励Ο繝・ヨ荳ｭ縺ｮ謫堺ｽ・

繧ｭ繝ｼ	蜍穂ｽ・
`Z`	繧ｺ繝ｼ繝☒繧､繝ｳ
`X`	繧ｺ繝ｼ繝☒繧｢繧ｦ繝・
`R`	譲呎ｺ悶え繧｣繝ｳ繝峨え
`T`	繝医Ξ繝ｼ繧ｹ ON/OFF
`B`	繝懊ャ繧ｯ繧ｹ繧ｺ繝ｼ繝☒
`+`	谺｡縺ｮ謨ｰ縺ｫ蛻・ｲ
`竊秦 / `竊蛋	繝医Ξ繝ｼ繧ｹ繧ｫ繝ｼ繧ｽ繝ｫ遘ｻ蜍・
`S`	BMP 菫晏ｭ・

8.4 G-Solve

- Root・育ｬｦ蜿ｷ螟牙喧繧ｹ繧ｭ繝｣繝ｳ + 莠悟・豕包ｼ・
- Maximum・域･ｵ螟ｧ・・
- Minimum・域･ｵ蟆擾ｼ・
- Intersection・・ 髱｢謨ｰ縺ｮ莠､轤ｹ・・
- y at x・・ 謖・ｮ壹〒 y 繧定ｨ育ｮ暦ｼ・
- x at y・・ 謖・ｮ壹〒 x 繧定ｨ育ｮ暦ｼ・
- Definite Integral・亥ｮ夂ｩ榊・・・

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9. Tools 繝「繝ｼ繝・

9.1 Spreadsheet・・5ﾃ・6・・

- 繧ｻ繝ｫ蜿ら・・啻A1`, `B3` 縺ｪ縺ｩ
- 譛ｬ蠑擾ｼ啻=A1+B1`, `=A1*2` 縺ｪ縺ｩ
- `Enter` 邱ｨ髮・/ `Del` 豸亥悉
- `S` CSV 菫晏ｭ・/ `L` CSV 隱ｭ縺ｿ霎ｼ縺ｿ
- `T` Sheet 廾・L1..L8 逶ｸ莠貞､画鋤

9.2 uLisp Interpreter

Scheme 鬚ｨ縺ｮ Lisp 繧ｵ繝悶そ繝・ヨ縲・

蜆ｩ逕ｨ蜿ｯ閭ｽ縺ｪ迚ｹ谿翫ヵ繧ｩ繝ｼ繝⊠・・

- `define`, `setq`, `lambda`, `let`
- `if`, `cond`, `and`, `or`, `not`
- `while`, `begin`, `progn`, `quote`

繝薙Ν繝医う繝ｳ髢｢譛ｬ・・

- 邂苓｡難ｼ啻+ - * / mod`
- 豈碑ｼ・ｼ啻< > <= >= =`
- 繝ｪ繧ｹ繝茨ｼ啻car cdr cons list null? atom? length append reverse nth`
- 鬮倬嚴・啻map filter reduce apply`
- 譛ｬ蟄ｦ・啻sin cos tan exp log sqrt abs floor ceil round expt pow`
- 邨ｱ險茨ｼ啻mean sum sd`
- 縺昴・莉厄ｼ啻print save load euler pi random`

**蝗ｳ蜉ｾ・*・・

```
(define (make-adder n) (lambda (x) (+ x n)))
(define add5 (make-adder 5))
(add5 3)   ; => 8

(map (lambda (x) (* x x)) '(1 2 3 4 5))   ; => (1 4 9 16 25)
(reduce + 0 '(1 2 3 4 5))                  ; => 15
```

**GC・医ぎ繝吶・繧ｸ繧ｳ繝ｬ繧ｯ繧ｷ繝ｧ繝ｳ・・*・・

- 繧ｻ繝ｫ繝励・繝ｫ・・500 蛟・
- 譏守､ｺ逧・`(gc)` 縺ｯ辟｡蜉ｹ蛹厄ｼ亥ｮ牙・譁ｹ縺ｮ縺溘ａ 蠖企勁・・
- 譛ｬ貂・凾・・*Tools 竊・uLisp 竊・Run GC now** 繧貞ｮ溯。・
- 縺ｾ縺溘・ **Reset interpreter**

9.3 Wi-Fi / Maxima

- SSID / Pass 險ｭ螳・
- 繝阪ャ繝医Ρ繝ｼ繧ｯ繧ｹ繧ｭ繝｣繝ｳ・域怙螟ｧ 30 莉ｶ・・
- 謗･邵・/ 蛻・妙 / 繧ｹ繝・・繧ｿ繧ｹ
- Maxima 驟｣謳ｺ・医・繝ｭ繝ｭ繧ｷ URL 縺ｯ蜷・・險ｭ螳夲ｼ・

9.4 Power / About

- 繝舌ャ繝・Μ繝ｼ谿矩醐繝ｻ蜈・崕迥ｶ諷・
- PSRAM 譛臥┌
- SD 繧ｫ繝ｼ繝臥憾諷・
- Wi-Fi 迥ｶ諷・
- Free heap
- `R`・售D 蜀崎ｵｷ陦・/ `T`・售D 繝・せ繝・

9.5 Settings・・VS 豌ｸ邵壼喧・・

- Angle: DEG / RAD / GRAD
- Display: Norm / Fix / Sci / Eng
- 譯∵焚・・繧・・・
- Norm 1 / 2 蛻・崛
- Save settings now
- Clear NVS

10. 險ｭ螳夲ｼ・VS 豌ｸ邵壼喧・・

險ｭ螳壹・荳肴尚逋ｺ諤ｧ繧ｹ繝医Ξ繝ｼ繧ｸ・・VS・峨↓菫晏ｭ倥＆繧後⊠∝・襍ｷ蜍墓ｙ後ｂ邵ｭ謖√＆繧後∪縺吶⊠・

螟画峩蠕後・ **Save settings now** 繧貞ｮ溯｡後@縺ｦ縺上□縺輔＞繧・

60 遘偵# 縺ｨ縺ｫ閾ｪ蜍穂ｿ晏ｭ倥b襍ｰ繧翫∪縺吶⊠・

11. 繝医Λ繝悶Ν繧ｷ繝･繝ｼ繝・ぅ繝ｳ繧ｰ

11.1 襍ｷ蜍輔＠縺ｪ縺・/ 逕ｻ髱｢「縺檎悄縺｣證・

- 繝舌ャ繝・Μ繝ｼ谿矩醐繧堤｢ｺ隱搾ｼ・SB 蜈・崕・・
- G0 繝懊ち繝ｳ + 繝ｪ繧ｻ繝・ヨ縺ｧ蜀崎ｵｷ蜍・
- USB 繧ｷ繝ｪ繧｢繝ｫ縺ｧ bootlog 繧堤｢ｺ隱搾ｼ・15200 bps・・

11.2 SD 繧ｫ繝ｼ繝峨′ 隱崎ｭ倥＆繧後↑縺・

- FAT32 蠖｢蠑上☒・2GB 莉･荳九ｒ菴ｿ逕ｨ
- 謗･轤ｹ繧呈ｸ・祉
- `Tools 竊・Power 竊・R` 縺ｧ繝ｪ繝医Λ繧､
- `Tools 竊・Power 竊・T` 縺ｧ隱ｭ縺ｿ譖ｸ縺阪ユ繧ｹ繝・

11.3 ﾎｼLisp 縺ｧ `ERR: unbound symbol`

- 繧ｻ繝ｫ繝励・繝ｫ謳ｯ貂・・蜿ｯ閭ｽ諤ｧ
- **Tools 竊・uLisp 竊・Run GC now**
- 縺ｾ縺溘・ **Reset interpreter**

11.4 Wi-Fi 縺ｫ謗･邯壹〒縺阪↑縺・

- 2.4GHz 蟶ｯ縺ｮ縺ｿ蟇ｾ蠢懶ｼ・GHz 荳榊庄・・
- SSID / Pass 繧貞・遒ｺ隱・
- 繝ｫ繝ｼ繧ｿ繝ｼ縺ｮ MAC 繝輔ぅ繝ｫ繧ｿ繧堤｢ｺ隱・

11.5 繧ｰ繝ｩ繝輔′ 讌募・縺ｫ縺ｪ繧・

- **Square aspect (1:1)** 繧帝∈謚・
- 縺ｾ縺溘・ Range 繧呈焔蜍戊ｪｿ謨ｴ

11.6 邨ｱ險医Ｍ繧ｹ繝医・荳企剔

- PSRAM 髢樊政霈峨・縺溘a 200 隕∫ｴ☒縺ｾ縺ｧ
- 999 隕∫ｴ☒縺ｯ PSRAM 謳ｭ霈峨ョ 繝舌う繧ｹ縺悟ｽ・ｦ・

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12. 莉倬鹸

12.1 繧ｭ繝ｼ謫ｳ隕玖。ｨ

繧ｭ繝ｼ	識溯・
`Fn + Esc`	譌ｻ繧・/ 繧ｭ繝｣繝ｳ繧ｻ繝ｫ
`Enter`	竪ｺ蝗・/ 蝗溯。・
`Del`	繝舌ャ繧ｯ繧ｹ繝壹・繧ｹ
`;`	荳・
`.`	荳・
`,`	蟾ｦ
`/`	蛇ｳ

12.2 邨・∩霎ｼ縺ｿ髢｢謨ｰ荳⊠隕ｧ・・alculator・・

`sin, cos, tan, asin, acos, atan, sinh, cosh, tanh, asinh, acosh, atanh, csc, sec, cot, csch, sech, coth, exp, ln, log, log10, log2, sqrt, cbrt, abs, sign, sgn, floor, ceil, round, int, frac, fact, gamma, lgamma, erf, erfc, deg, rad, isPrime, prime, nextPrime, prevPrime, random, randInt, rint, randN, toF, toC, toK, pow, nrt, atan2, hypot, min, max, mod, gcd, lcm, nPr, nCr, logn, randR, if, clamp, dms`

12.3 蝗壽焚

- `pi` 窶・蜀・捉邇・
- `e` 窶・繧ｪ繧､繝ｩ繝ｼ謨ｰ
- `euler` 窶・繧ｪ繧､繝ｩ繝ｼ謨ｰ・医お繧､繝ｪ繧｢繧ｹ・・
- `phi` 窶・鮟・≡豈・
- `gamma` 窶・繧ｪ繧､繝ｩ繝ｼ繝ｻ繝槭せ繧ｱ繝ｭ繝ｼ繝句ｮ壽焚
- `inf` 窶・辟。髯仙､ｧ
- `nan` 窶・NaN
- `ans` 窶・逶ｴ蜑阪・邨先棡
- `mem` 窶・繝。繝｢繝ｪ M

12.4 螟画焚

- `A`繧象J` 窶・繝ｦ繝ｼ繧ｶ繝ｼ螟画焚・亥､ｧ譁・ｭ暦ｼ・
- `a`繧象j` 窶・繝ｦ繝ｼ繧ｶ繝ｼ螟画焚・亥ｰ乗枚蟄暦ｼ・
- `E` 窶・螟画焚 E・・0.92 莉･髯搾ｼ・
- `x`, `t`, `n` 窶・謨ｰ蠑丞・縺ｮ莠育ｴ・､画焚・域悴蝗ｺ譛ｬｯ譎・0・・

PoMaS v0.92 寠・Paper UI v2

Author: jpeqtx